

# SRE Dynamics Supplementary Note:

Magnitude Anchoring of Logical Depth  $N$  and Observational-Protocol Safety Masking

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**Version:** v1.6 (Axiom-Homomorphic-Mapping Aligned Revision)

Version Note: This supplementary note is aligned with the full SRE v1.6 manuscript suite.  $l_{min}$  denotes the **minimal instantiable topological scale (emergent ontological ultraviolet boundary)** of the rendering layer. The electron is a self-consistent topological closed-loop that emerges in the physical rendering-layer after multi-scale rigid-boundary screening and homomorphic mapping. The safety discussed herein applies to deep internal topological degrees-of-freedom of the closed loop; routine physical manipulation of the electron as a whole object within the rendering layer is not prohibited.

## 1. Derivation of Logical Depth $N$

Within the SRE framework, the electron is a stable self-consistent topological closed-loop residing in the physical rendering layer. The parameter  $N$  represents the equivalent count of instantiable topological scales for this closed loop. It is not an empirical constant native to the underlying causal space, but jointly determined by the characteristic rendering scale of the system and the lower instantiable-scale bound of the rendering layer.

- Formula:

$$N = \lambda_c / l_{min}$$

- Magnitude evaluation: Substituting known parameters (Compton wavelength  $\lambda_c \approx 10^{-12}$  m and minimal instantiable topological scale  $l_{min} \approx 10^{-35}$  m), the equivalent logical depth of the electron yields:

$$N \approx \frac{10^{-12}}{10^{-35}} = 10^{23}$$

Logical conclusion: The electron is not a simple primitive topological unit, but a high-order emergent topological module with extremely high redundancy (equivalent topological count of order  $10^{23}$ ).

## 2. Sampling Theorem and Observational Masking

This supplementary note emphasises that the “invisibility” of deep internal topological details inside the electron closed-loop is enforced jointly by information-theoretic sampling limits and coarse-graining effects of multi-scale homomorphic mapping.

- Resolution gap: All physical observational protocols are bounded by the rendering-layer minimal instantiable topological scale  $l_{min}$ . The resolving power of a single physical observation for the full internal topology of the electron closed-loop is merely on the order of  $\frac{1}{N}$  (i.e.  $1/10^{23}$ ).
- Statistical smoothing: Constrained by multi-scale rigid-boundary filtering and full homomorphic-mapping mechanisms, numerous fine-grained internal topological degrees-of-freedom of the closed-loop undergo many-to-one coarse-grained compression. Only statistically stable collective effects are preserved and finally manifest as observable mass, charge and spin within the physical rendering layer.

### 3. Core Safety Criterion: Strict Read-Only Physical-Level Restriction

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The core significance of this supplementary note is to demonstrate the intrinsic physical safety within this theoretical model.

- **Permission Lock:**  $l_{min}$  constitutes the lower scale bound for reciprocal-measurement addressing inside the physical rendering layer. Any attempt to directly intervene inside the electron closed-loop and perform operations at equivalent scales far smaller than  $l_{min}$  cannot achieve valid physical addressing.
- **Causal Barrier:** A vast order-of-magnitude gap exists between the electron's equivalent logical depth  $N$  and physical observational limits. Deep internal topological degrees-of-freedom of the closed-loop are naturally isolated within a physically-inaccessible black box.

Important distinction: This restriction **does not forbid routine physical manipulation of the complete electron closed-loop within the rendering-layer**, such as electromagnetic coupling or particle-scattering experiments. What is prohibited is penetrating the closed-loop to rewrite its deep internal topological details.

- **Conclusion:** Observers may only read stable observables delivered by homomorphic-mapping in the rendering layer; no instructions can be written to the deep internal topology of the closed-loop. Constraints originating from instantiation boundaries form an intrinsic cosmic-scale hard-encryption barrier.

### 4. Final Conclusion

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This supplementary note establishes the following inference: physical reality is the rendering output after homomorphic-mapping sampling of causal space.

The enormous equivalent topological count ( $10^{23}$ ) of the electron closed-loop safeguards the stability of particle-based matter. Subject to natural constraints imposed by rendering-layer instantiation boundaries, deep internal topological degrees-of-freedom of this topological closed-loop are physically inaccessible and unmodifiable, possessing intrinsic safety properties.

**Author's Note:** Completion of this supplementary note marks that SRE-Dynamics has not only achieved a closed deductive chain spanning micro- to macro-scales at the logical level, but also constructed a theoretically impenetrable protective boundary of physical origin.

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